E-mail campaign test: send the men's e-mail to all customers

To: Marketing / CRM leads · From: Jing Tong, Data Analyst · Data: Hillstrom e-mail experiment, 64,000 customers, 2-week outcome window

Bottom line. Both e-mails increase visits, purchases and spend, but the **men's e-mail is clearly better** and works in every customer segment. Sending it to everyone yields about **+$221 profit per 1,000 customers** compared with no e-mail. Tailoring the e-mail to customer segments did **not** add measurable value, so the simple rule is recommended.

WHAT WE TESTED

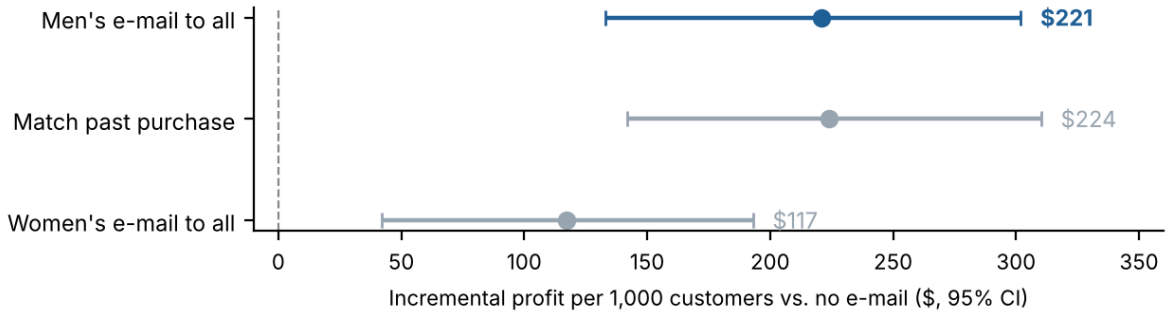
Customers who bought in the last 12 months were randomly split into three equal groups: men's e-mail, women's e-mail, or no e-mail. Randomization checks passed (group sizes p = 0.90; all pre-campaign characteristics balanced, max |SMD| = 0.014), so differences can be attributed to the e-mails.

RESULTS VS. NO E-MAIL

Metric	No e-mail	Men's e-mail	Women's e-mail
Visit rate	10.6%	**18.3% (+7.7 pp)**	15.1% (+4.5 pp)
Conversion rate	0.57%	**1.25% (+0.68 pp)**	0.88% (+0.31 pp)
Spend per customer	$0.65	**$1.42 (+$0.77)**	$1.08 (+$0.42)
Incremental profit per 1,000 customers*	–	**+$221 [133, 302]**	+$117 [42, 193]

All lifts significant after Holm correction ($p < 0.01$). Brackets: 95% bootstrap CI. *Assumes 30% gross margin and $0.01 per e-mail.

SHOULD WE TARGET DIFFERENT CUSTOMERS WITH DIFFERENT E-MAILS?



- **Match past purchase** (women-only buyers get the women's e-mail): +$3 per 1,000 vs. men's-to-all, CI [−51, +61], i.e. no detectable difference.
- The men's e-mail raised spend in every segment; the women's e-mail showed no clear effect for men-only buyers, existing customers and phone customers.
- A **rule learned from the data** looked better in-sample (+$19) but **lost $27 per 1,000** on held-out customers and won in only 26% of 200 train/test splits: it was fitting noise.
- Conclusion: extra targeting complexity is not justified by this data.

HOW ROBUST IS THIS?

- Sending the men's e-mail breaks even up to **~$0.23 per e-mail** (30% margin) and stays profitable for every tested margin (20–50%) and cost ($0–0.10) combination.
- Variance-reduction (CUPED) and covariate adjustment left all estimates unchanged.

CAVEATS & NEXT STEPS

- **Not measured:** unsubscribes, e-mail fatigue and effects beyond two weeks. Future campaigns should track unsubscribe rate as a guardrail metric.
- **Follow-up test:** the one promising segment signal, the women's e-mail working for *new* customers, is exploratory (unadjusted p ≈ 0.02). Test men's vs. women's e-mail among new customers with ~26,000 per arm (detects a 0.2 pp conversion difference at 80% power).

Methods: SRM χ^2 test, standardized mean differences, two-proportion z-tests with Holm correction, bootstrap CIs for spend, CUPED, power/MDE analysis, inverse-probability-weighted policy evaluation with repeated train/test splits. Code: github.com/JingTong01/email-campaign-ab-test